GYANMANJARI INNOVATIVE UNIVERSITY

Gyanmanjari Institute of Technology – Faculty of Engineering &Technology

Business Intelligence and Data Governance Case Study

RetailMart: OLAP-based decision making, data quality, metadata, privacy and ethics in data mining

Student Name	Kothari Tarj Mehulbhai	Enrollment No.	260205020001
Subject	DataWarehouseandDataMining (METCE11507)	Semester	1 (M.Tech)
Activity	ALA 3	Marks	10

1. Aim and Scenario

To analyse a real-world type of scenario involving **data warehouse usage, OLAP-based decision making and organizational data governance**, covering **data quality challenges, metadata management and ethical concerns in data mining**, and to propose solutions and **governance guidelines**.

Scenario. RetailMart (the fictitious retail chain of ALA 1; the data are simulated) has built a data warehouse and wants managers to use OLAP reports for stocking, promotion and store decisions. The BI teamalsowantstominecustomerpurchasesforpersonalisedoffers.Managementasksthreequestions:
(1) What do the OLAP results tell us? (2) Can we trust the numbers? (3) How can the company use customer data responsibly and legally? Real incidents (Target, Netflix Prize) and India's data protection law are used to ground the analysis.

2. Business Intelligence: OLAP-based Decisions

Results below come from queries on the warehouse (see ALA 1). Each decision uses an OLAP operation and carries a caveat that links BI to data quality.

OLAP finding	Operation	Decision / recommendation
Netsalesgrew34%(Rs.3.67Min2024to Rs.4.94Min2025).Electronicsgave**87%** of Q4-2025 revenue; laptops and smartphonesaloneare83%ofallrevenue.	Roll-up;slice(Q4 2025)	**Concentration risk:** keep safety stock and supplier contracts for these two products, track stock-outs and price changes weekly, and developgrocery/personal-carefootfalltoreduce dependence.
The West region gives 67% of Electronics sales;thetopstoreisAhmedabadMall(Rs. 1.37 M).	Dice (category × region),roll-upto store	Allocate premium electronics stock to West stores;studywhySouth/Eastunder-perform before investing.
Average daily sales in the festive months (Oct–Nov)areonly**3%**aboveregulardays although discounts are deeper.	Slicebyseason (dim_date)	Reviewthefestivediscountpolicy:theuplift may not repay the discount. Add **cost and margin** data to the warehouse (currently missing) before judging profitability; run controlled promotion tests.
UPI carries 46% of sales value.	Roll-upbypayment mode	Offer UPI-linked loyalty rewards; negotiate paymentcharges;keepcashhandlingforthe rest.

Caution: OLAP answers are only as reliable as the data behind them. Section 3 shows how much the numbers could have been distorted without data-quality controls.

3. Data Quality Challenges

The raw extracts contained typical quality problems. The table shows what was found by profiling the source files and how it was handled in the ETL (ALA 1).

Quality dimension	Problem found in the source data	Treatment (ETL rule)
Uniqueness	105 exact duplicate sales lines (1.95%); 12 duplicate customer records (2.9%)	Removeduplicates;enforceprimary key (invoice_id, line_no)
Completeness	19customerswithoutcity;0.8%oflineswithoutprice; 1.1% without quantity	City = 'Unknown'; price from product master;lineswithoutquantityrejected
Validity	13invalide-mailaddresses;futureorderdates; discounts above 50%; negative quantities	Flag e-mail; reject rows that violate businessrules(loggedinetl_rejects)
Consistency	28.8%ofdatesnotinoneformat;pricesstoredastext (2.9%); 11 spellings of 4 payment modes; 8 gender codes; 12 state spellings	Standardiseformatsandcodelists (reference/master data)
Referential integrity	Customeridsinsalesthatdonotexistinthe customer master (0.85% of lines)	Mapto'Unknown'member(key0)so facts are not lost; report to the data owner

Business impact (measured): if the 105 duplicate lines had not been removed, total net sales would have been overstated by about **Rs. 247,677 (2.88%)** (the clean total is Rs. 8,613,661), mainly because duplicated laptop sales are so valuable. 127 lines were rejected by business rules, and 0.85% of lines (0.66% of value) cannot be linked to a known customer, which weakens customer-level analysis. After the ETL: 5,153 fact rows, 0 foreign-key orphans, 12 flagged e-mails and 18 'Unknown' cities remain for correction at the source.

4. Metadata Management

Metadata is 'data about data'. Without it, two managers may compute 'sales' differently and nobody can trace a number to its source.

Type	Content	Example from the RetailMart warehouse
Business metadata	Definitions,owner,rules, hierarchies	**net_amount**=quantity×unitprice−discount,excludingtax; owner: Head of Sales; hierarchy date: day →month →quarter →year
Technical metadata	Tables,columns,types,keys, schemas	fact_sales(date_keyINTEGER,…,net_amountREAL); surrogate keys; star schema diagram
Operational metadata	ETLrunstatistics,errors,load times	Runlog:5,385read;105duplicatesremoved;127rejected (table etl_rejects); 5,153 loaded
Lineage	Whereavaluecamefromand how it was changed	fact_sales.net_amount←sales_raw.quantity,unit_price(text cleaned, filled from products.csv), discount_pct ←rules: positive quantity, discount ≤50%

Proposal: maintain a data dictionary and business glossary (one definition per KPI), store ETL logs permanently, and publish lineage for every dashboard measure in a data catalogue.

5. Privacy, Ethics and Governance

Ethical concerns in data mining: two real cases

Case	What happened (as widely reported)	Lesson for RetailMart
Targetpregnancy prediction (2012)	AUSretailerbuiltascorefromshoppingbasketsto predict customers' pregnancy and target offers; a New York Times Magazine article (Duhigg, 2012) described how such inference could reveal a sensitive condition to the customer's family.	Association/classification on baskets can **infer sensitive facts** (health, familystatus)thatthecustomernever disclosed. Purpose limitation and sensitivity checks are required.
NetflixPrizedataset (2006–2010)	Netflix released about 100 million 'anonymised' movie ratings for a contest. Narayanan and Shmatikov (2008) showed that individuals could be re-identified by matching a few ratings with public IMDbdata.Netflixlatercancelledaplannedsecond contest.	**Removing names is not anonymisation.** Combinations of quasi-identifierscanidentifypeople; test re-identification risk before sharing data.

Measured re-identification risk in the customer dimension (k-anonymity)

A data set is **k-anonymous** if every record shares its combination of quasi-identifiers with at least k−1 other records (Sweeney, 2002). The customer dimension (400 known customers) was tested:

Quasi-identifiers used	Groups	Smallestgroup (k)	Uniquerecords (k = 1)	Recordsin groups <5
city	9	18	0	0.0%
gender + age_group + city	69	1	5	16.5%
gender + age_group + city + signup_year	233	1	121	96.2%
gender + age_group + state	46	1	1	7.8%
age_group + state	24	2	0	4.5%

With only gender, age group and city, 5 customers are already unique and 16.5% sit in groups smaller than 5; adding signup year makes 121 customers unique. A report that shows such combinations could identify individuals. **Generalising** the data (age group + state) raises the smallest group to k = 2 and reduces risk; suppression of small cells (k < 5) should be added for published reports.

Pseudonymisation of customer identifiers

Analysts do not need real customer ids. A salted hash gives a stable pseudonym that still allows joining and counting (the salt is a secret kept in a key vault):

customer_id	Name (restricted)	Age group	City	Pseudonym for analytics
C0072	Isha Singh	26-35	Mumbai	CUST_a5c16fed4b
C0390	Aarav Verma	51+	Bhavnagar	CUST_d1c246175e
C0261	Yash Reddy	51+	Mumbai	CUST_7dfa096fe5
C0084	Ananya Kumar	51+	Mumbai	CUST_0c828741e1
C0036	Pooja Nair	51+	Ahmedabad	CUST_2a43916370

```
importhashlib
def pseudo(customer_id, salt):                # salt = secret value stored outside the database
    return "CUST_" + hashlib.sha256((salt + customer_id).encode()).hexdigest()[:10]
```

Legal context in India

- The **Digital Personal Data Protection (DPDP) Act, 2023** was enacted on 11 August 2023, and the **DPDP Rules, 2025** were notified on 13 November 2025 with a phased start: the Data Protection Board provisions at once, consent-manager obligations after 12 months and the main obligations of data fiduciaries after 18 months (by 13 May 2027).
- Main duties of a data fiduciary such as RetailMart: a clear notice and valid consent for each purpose, use of data only for that purpose, security safeguards, breach intimation, limits on retention and erasure, rights of the data principal to access and correct data, and verifiable consent for children's data.
- *This is a summary for study purposes, not legal advice; the official Act and Rules should be consulted for exact requirements.*

6. Proposed Solutions and Governance Guidelines

Data governance is the set of roles, policies, standards and processes that make data accurate, secure, usable and compliant. Proposed roles: **Data Owner** (accountable for a data domain, e.g. Head of Sales), **Data Steward** (day-to-day quality and definitions, BI analyst), **Data Custodian** (IT/DBA: storage, access, backup) and a **Data Protection Officer** (privacy and legal compliance).

#	Guideline	Control / action	Measure (KPI)
G1	Singledefinitionof every KPI	Businessglossaryanddatadictionary approved by owners	100%ofdashboardKPIshavea definition and owner
G2	QualitygatesintheETL	Validationrules,rejecttable,row-count reconciliation for each load	Duplicates in warehouse = 0; rejectedlines<2%(now2.4%)
G3	Fixproblemsatthe source	Mandatoryfieldsandformatvalidationatthe billing counter; unique invoice numbers	Invalide-mails<1%;orphan customer ids = 0
G4	Master data management	Onecustomermaster;controlledcodelists (gender, state, payment mode)	Spelling variants = 0
G5	Data classification	Label columns: public, internal, confidential, personal/sensitive(name,e-mail,birthdate= personal)	All warehouse columns classified
G6	Least-privilege access	Role-based access; analysts see pseudonymisedcustomers;DBAsaccess logged	Quarterlyaccessreview completed
G7	Anonymisationbefore sharing	Generalisequasi-identifiers;suppresscells with k < 5; test k-anonymity	k ≥5 for every published report
G8	Purposelimitationand ethics review	Revieweachminingproject(e.g.personalised offers); avoid inferring sensitive traits; no use of health-related baskets for targeting without consent	100% of mining projects reviewed
G9	Consentand transparency	Plain-languagenotice,purpose-wiseconsent, easy withdrawal, grievance contact	Consentrecordsforallloyalty customers
G10	Retention and erasure	Deleteoranonymisepersonaldataafterthe retention period or on request	Erasurerequestsclosedwithin the policy time
G11	Securityandbreach response	Encryption at rest and in transit, backups, testedincidentplanwithtime-boundreporting	Breach drill every year
G12	Lineage and audit	ETL logs and lineage kept; periodic data-quality report to the governance council	Monthly quality score published

Phase	Actions
0–3 months	Appointowners,stewardsandDPO;buildglossary;addETLqualitygatesandrejectreporting; classify data; pseudonymise customer ids in the analytics layer.
3–6 months	Fixsource-systemvalidation;masterdataforcustomersandcodelists;role-basedaccess; consent notice and records; k-anonymity checks in report publishing.
6–12 months	Datacataloguewithlineage;ethicsreviewboardforminingprojects;breachdrill;retention/erasure automation; align fully with the DPDP Rules before the 2027 deadline.

7. Conclusion

Thewarehousegaveclear,decision-readyinsights(revenueconcentrationinafewelectronicsproducts,a modest festive uplift, regional differences), but the case study shows that BI is only trustworthy when data quality is managed: skipping de-duplication alone would have overstated sales by 2.9%. Metadata and lineage make the numbers explainable. Mining customer data also brings ethical and legal duties: the Target and Netflix cases and the measured k-anonymity results show that sensitive facts can be inferred and that 'anonymised' data can still identify people. The proposed governance guidelines (quality gates, master data, classification, least privilege, anonymisation, purpose limitation, consent, retention, security and audit) give RetailMart a practical route to using its data responsibly and in line with the DPDP Act and Rules.

References

- JiaweiHan,MichelineKamberandJianPei,*DataMining:ConceptsandTechniques*,Morgan Kaufmann Publishers (data quality, privacy and security in data mining).
- Alex Berson and Stephen J. Smith, *Data Warehousing, Data Mining and OLAP*, Tata McGraw-Hill.
- DAMAInternational,*DAMA-DMBOK:DataManagementBodyofKnowledge*,2nded.,2017(data governance, metadata, data quality).
- A.NarayananandV.Shmatikov,"Robustde-anonymizationoflargesparsedatasets",*Proc.IEEE Symposium on Security and Privacy*, 2008.
- L.Sweeney,"k-anonymity:amodelforprotectingprivacy",*InternationalJournalofUncertainty,Fuzziness and Knowledge-Based Systems*, 10(5), 2002.
- C. Duhigg, "How companies learn your secrets", *The New York Times Magazine*, 2012.
- GovernmentofIndia,MinistryofElectronicsandInformationTechnology:DigitalPersonalDataProtection Act, 2023, and Digital Personal Data Protection Rules, 2025.